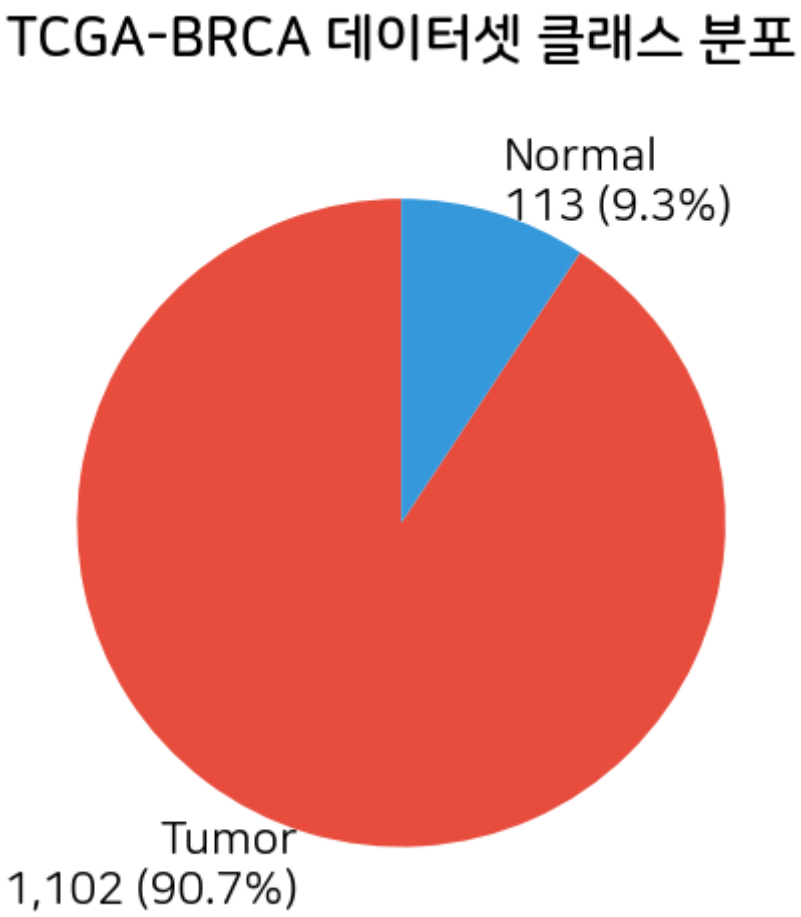
# Topological Autoencoder with COAST Loss

Experiment Results & Analysis

TCGA-BRCA Breast Cancer Classification | Lucius Hwang | 2026-04-01

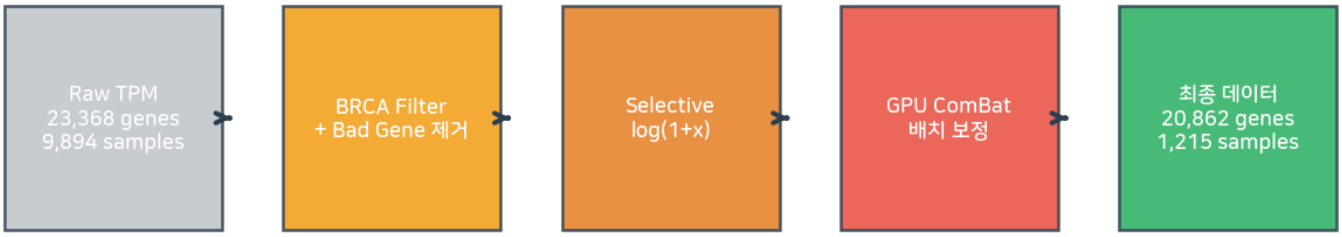
## 1. Dataset




| 항목 | 값 |
|---|---|
| 데이터 소스 | GEO GSE62944 (TCGA RNA-seq, Rsubread TPM) |
| 암종 | TCGA-BRCA (Breast Invasive Carcinoma) |
| 전체 샘플 수 | **1,215** (Tumor 1,102 / Normal 113) |
| 원본 유전자 수 | ~23,368 |
| 최종 유전자 수 (필터링 후) | **20,862** |
| 제거된 Bad Genes | 2,068개 (\|skewness\|>2.0 또는 excess kurtosis>10.0) |
| 배치 소스 | 31개 BRCA TSS (Tissue Source Site) |
| 발현 단위 | TPM (Transcripts Per Million) |

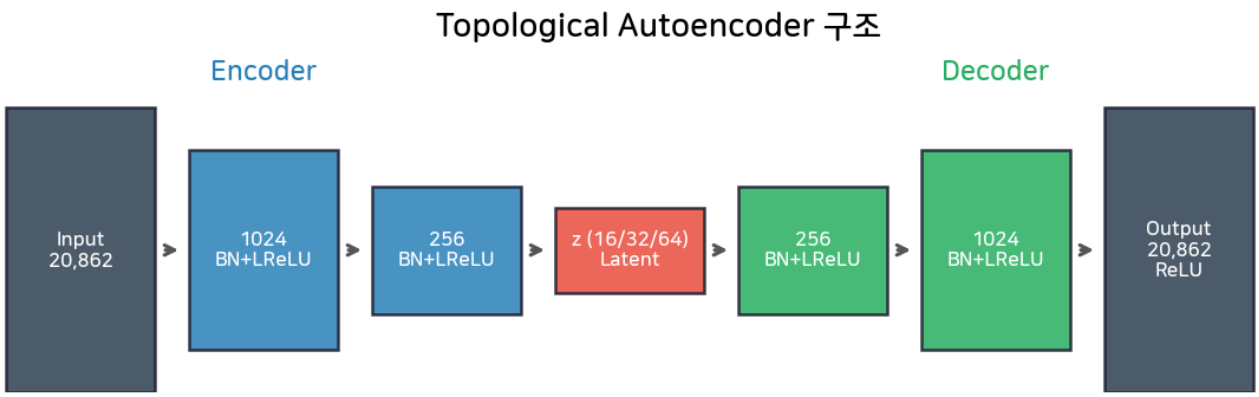
## 2. Preprocessing Pipeline

### 전처리 파이프라인




| 단계 | 처리 내용 | 비고 |
|---|---|---|
| 1. 데이터 병합 | Tumor + Normal TPM 합치기, BRCA 환자 필터링 (31개 TSS) | 9,894 → 1,215 samples |
| 2. Bad Gene 제거 | overall_bad_genes_4fold.txt 기반 제거 | 2,068 genes 제거 |
| 3. Log 변환 | \|skew\|>2.0 또는 kurtosis>10.0인 유전자에 log(1+x) | 청크(1000 genes) 처리 |
| 4. 배치 보정 | GPU ComBat (Empirical Bayes + Location-Scale) | CPU 15분 → GPU 2.8초 |
| 5. 최종 출력 | cleaned_tcga_tpm_for_TAE.csv | 1,215 × 20,862 |

## 3. Model Architecture




| Encoder | | Decoder | |
|---|---|---|---|
| Layer | Details | Layer | Details |
| Linear | 20,862 → 1,024 | Linear | latent_dim → 256 |
| BatchNorm1d | + LeakyReLU(0.2) | BatchNorm1d | + LeakyReLU(0.2) |
| Linear | 1,024 → 256 | Linear | 256 → 1,024 |
| BatchNorm1d | + LeakyReLU(0.2) | BatchNorm1d | + LeakyReLU(0.2) |
| Linear | 256 → latent_dim | Linear | 1,024 → 20,862 |
| No activation (full $\mathbb{R}$) | | ReLU (TPM ≥ 0) | |

# 4. Model Configurations

## 4.1 거리 함수 (Distance Metrics)

| 거리 함수 | 수식 | 특성 |
|---|---|---|
| Euclidean | $D = \sqrt{\Sigma(x_i - y_i)^2}$ | 절대 거리; 고차원에서 수렴 (curse of dimensionality) |
| Cosine | $D = 1 - (x \cdot y)/(\lVert x \rVert \cdot \lVert y \rVert)$ | 각도 차이; 크기 불변 (magnitude-invariant) |
| Pearson | $D = 1 - r$ (Pearson corr.) | 상관 거리; 스케일+시프트 불변 |

## 4.2 손실 함수 (Loss Functions)

| 구분 | Baseline (MSE) | Proposed (COAST) |
|---|---|---|
| 위상 손실 | MSE(D̃_latent, D̃_original) | Debiased Sinkhorn Divergence |
| 거리 함수 | Euclidean | Cosine (Optimized) |
| 비교 방식 | 거리 행렬 원소별 비교 | 거리 분포 간 최적 수송 비용 |
| 가중치 | 고정 $\lambda = 1.0$ | Adaptive (학습 가능, Kendall 2018) |
| 이론적 근거 | 유클리드 근사 | Wasserstein 거리의 근사 |

## 4.3 COAST Loss 핵심 수식

```
거리 → 확률: P_ij = exp(-D²_ij / 2σ²) / Σ_k exp(-D²_ik / 2σ²)

엔트로피 정규화 OT: OT_ε(a,b) = min_π ⟨C,π⟩ + ε·KL(π ‖ a⊗b)

비편향 발산: S(a,b) = OT(a,b) - ½·OT(a,a) - ½·OT(b,b) ≥ 0

COAST Loss: L = ½·e^(-s_r)·L_recon + ½·e^(-s_t)·m·L_topo + ½·s_r + ½·s_t
```

COAST = Cosine Optimized Adaptive Sinkhorn Transport
구현 트릭: Envelope Theorem (역전파 VRAM ~50× 절감), Log-domain Sinkhorn (수치 안정성), 학습 가능 σ (중앙값 거리 초기화)

## 5. Training Configuration

| Parameter | Value |
|---|---|
| Optimizer | Adam (lr=1e-4, weight_decay=1e-5) |
| Batch Size | 64 |
| Epochs | 100 |
| Validation Split | 20% (계층화 분할) |
| Topo Weight (baseline) | λ = 1.0 |
| 분류기 평가 | LogisticRegression (10 에포크마다) |
| Best Model 선택 | 최소 검증 총 손실 |
| GPU | NVIDIA RTX 4060 (8GB VRAM) |

### 실험 그리드 (총 27개 구성)

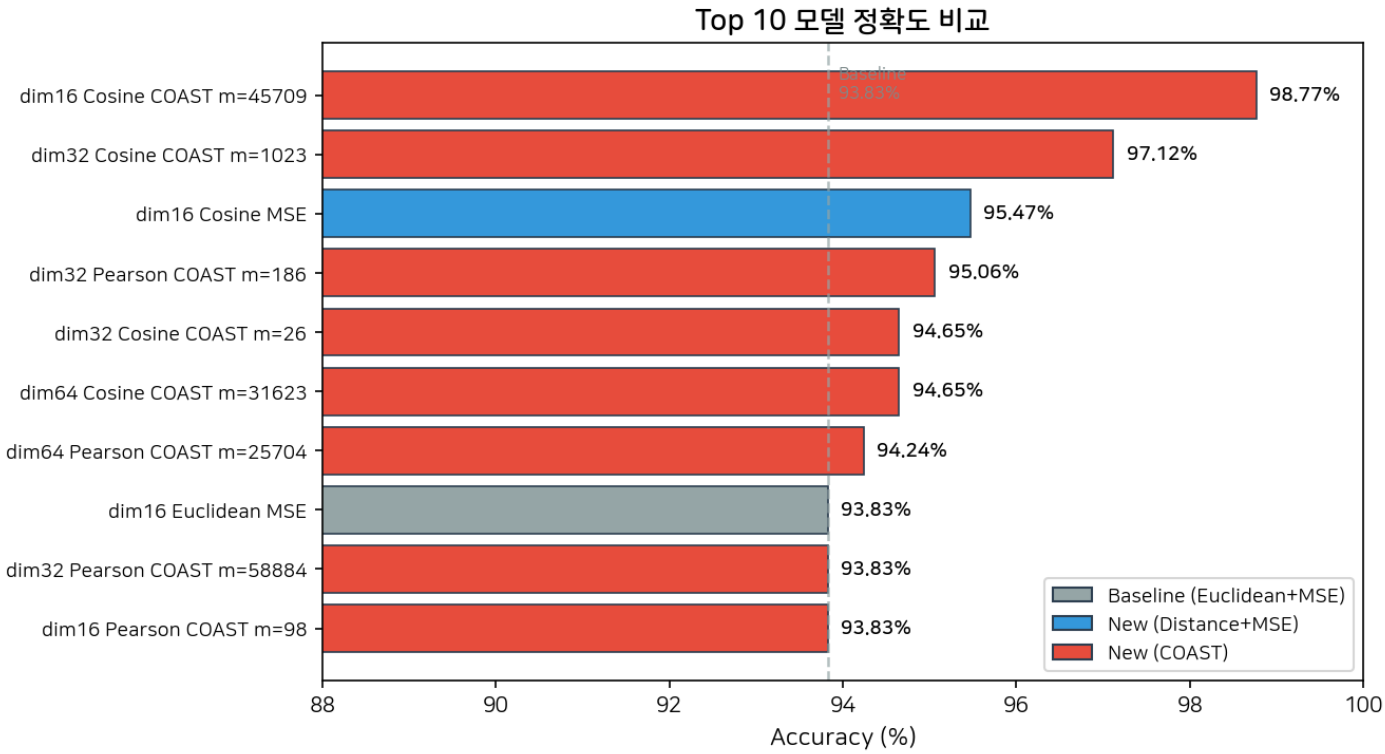
| Factor | Values | Count |
|---|---|---|
| 잠재 차원 | 16, 32, 64 | 3 |
| 거리 함수 | Euclidean, Cosine, Pearson | 3 |
| 손실 함수 | Standard MSE, COAST | 2 |
| Topo Multiplier | 3 values per config (log-scale sweep) | 3 |

9 baseline (3 metrics × 3 dims) + 18 COAST (2 metrics × 3 dims × 3 multipliers) = 27 total

## 6. Main Results

Best: dim16 + COAST (m=45,709) → Accuracy 98.77%
Baseline (dim16 + Euclidean + MSE) 93.83% 대비 +4.94%p 향상

### 6.1 Top 10 모델 정확도



### 6.2 전체 27개 구성 결과표

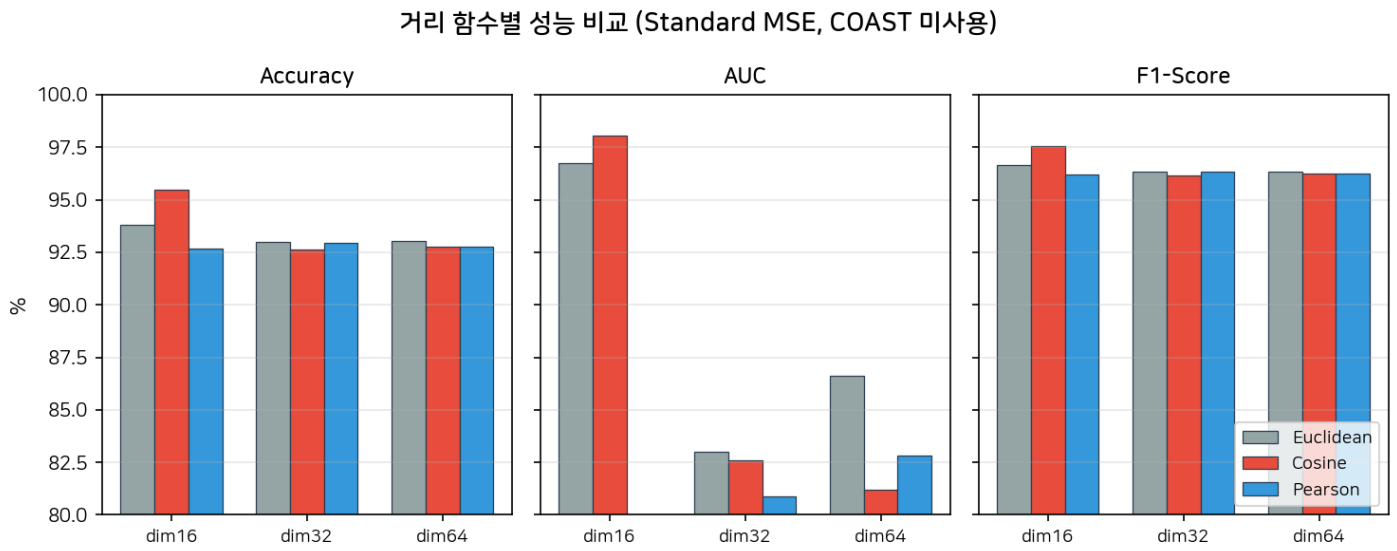
| # | Dim | Distance | Loss | Topo m | Accuracy | AUC | F1 | Prec. | Recall |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 16 | Cosine | COAST | 45709 | **98.77%** | 0.988 | 0.993 | 0.991 | 0.995 |
| 2 | 32 | Cosine | COAST | 1023 | **97.12%** | 0.989 | 0.984 | 0.986 | 0.982 |
| 3 | 16 | Cosine | Std MSE | — | **95.47%** | 0.980 | 0.975 | 0.960 | 0.991 |
| 4 | 32 | Pearson | COAST | 186 | **95.06%** | 0.986 | 0.973 | 0.964 | 0.982 |
| 5 | 32 | Cosine | COAST | 26 | **94.65%** | 0.972 | 0.971 | 0.952 | 0.991 |
| 6 | 64 | Cosine | COAST | 31623 | **94.65%** | 0.972 | 0.971 | 0.960 | 0.982 |
| 7 | 64 | Pearson | COAST | 25704 | **94.24%** | 0.981 | 0.968 | 0.960 | 0.977 |
| 8 | 16 | Euclidean | Std MSE | — | **93.83%** | 0.968 | 0.967 | 0.944 | 0.991 |
| 9 | 32 | Pearson | COAST | 58884 | **93.83%** | 0.980 | 0.966 | 0.956 | 0.977 |
| 10 | 16 | Pearson | COAST | 98 | **93.83%** | 0.949 | 0.967 | 0.944 | 0.991 |
| 11 | 64 | Euclidean | Std MSE | — | **93.03%** | 0.866 | 0.964 | 0.934 | 0.995 |
| 12 | 16 | Euclidean | Std MSE | — | **92.98%** | 0.797 | 0.963 | 0.930 | 0.999 |
| 13 | 32 | Euclidean | Std MSE | — | **92.98%** | 0.830 | 0.963 | 0.930 | 0.999 |
| 14 | 32 | Pearson | Std MSE | — | **92.93%** | 0.809 | 0.963 | 0.930 | 0.999 |
| 15 | 64 | Pearson | Std MSE | — | **92.77%** | 0.828 | 0.962 | 0.929 | 0.998 |
| 16 | 64 | Cosine | Std MSE | — | **92.77%** | 0.812 | 0.962 | 0.929 | 0.998 |
| 17 | 16 | Pearson | Std MSE | — | **92.67%** | 0.790 | 0.962 | 0.927 | 1.000 |
| 18 | 32 | Cosine | Std MSE | — | **92.62%** | 0.826 | 0.962 | 0.927 | 0.998 |
| 19 | 16 | Cosine | Std MSE | — | **92.52%** | 0.800 | 0.961 | 0.926 | 0.999 |
| 20 | 16 | Cosine | COAST | 32 | **91.77%** | 0.968 | 0.956 | 0.920 | 0.995 |
| 21 | 64 | Cosine | COAST | 6607 | **91.77%** | 0.965 | 0.957 | 0.917 | 1.000 |
| 22 | 64 | Cosine | COAST | 1622 | **91.77%** | 0.966 | 0.957 | 0.917 | 1.000 |
| 23 | 64 | Euclidean | Std MSE | — | **91.36%** | 0.976 | 0.954 | 0.920 | 0.991 |
| 24 | 64 | Pearson | COAST | 11 | **91.36%** | 0.924 | 0.954 | 0.913 | 1.000 |
| 25 | 16 | Pearson | COAST | 13183 | **90.53%** * | 0.809 | 0.950 | 0.905 | 1.000 |
| 26 | 16 | Pearson | COAST | 6918 | **90.53%** * | 0.887 | 0.950 | 0.905 | 1.000 |
| 27 | 16 | Pearson | Std MSE | — | **90.53%** * | 0.863 | 0.950 | 0.905 | 1.000 |
| 28 | 32 | Cosine | COAST | 3802 | **90.53%** * | 0.877 | 0.950 | 0.905 | 1.000 |
| 29 | 32 | Cosine | Std MSE | — | **90.53%** * | 0.885 | 0.950 | 0.905 | 1.000 |
| 30 | 64 | Pearson | Std MSE | — | **90.53%** * | 0.972 | 0.950 | 0.905 | 1.000 |
| 31 | 64 | Pearson | COAST | 55 | **90.53%** * | 0.932 | 0.950 | 0.905 | 1.000 |
| 32 | 64 | Cosine | Std MSE | — | **90.53%** * | 0.908 | 0.950 | 0.905 | 1.000 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 33 | 16 | Cosine | COAST | 186 | **90.12%** | 0.934 | 0.947 | 0.926 | 0.968 |
| 34 | 32 | Euclidean | Std MSE | — | **89.71%** | 0.957 | 0.945 | 0.911 | 0.982 |
| 35 | 32 | Pearson | Std MSE | — | **89.71%** | 0.958 | 0.945 | 0.911 | 0.982 |
| 36 | 32 | Pearson | COAST | 12589 | **89.30%** | 0.924 | 0.943 | 0.911 | 0.977 |

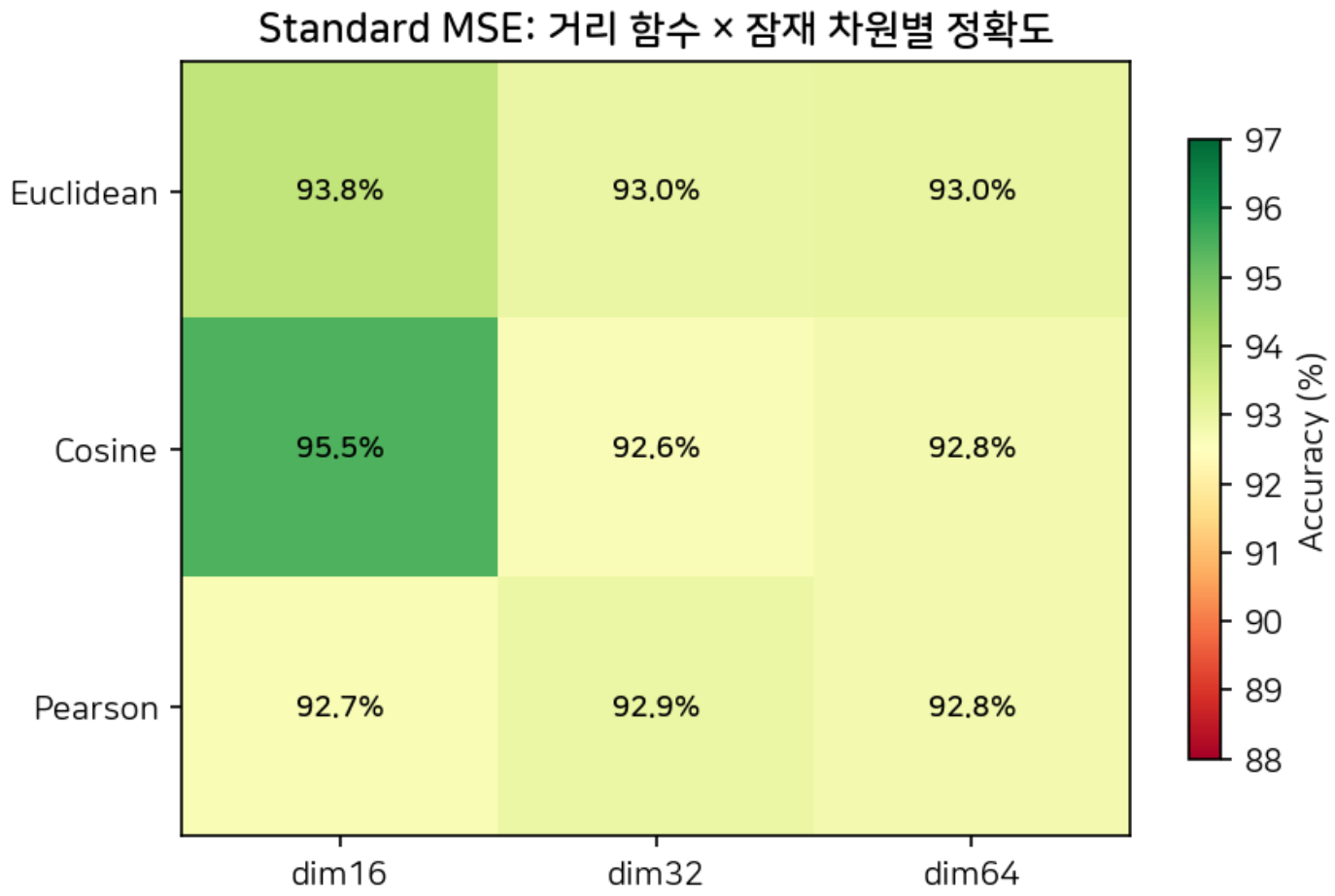
노란색: Best model | 회색: Baseline | * 표시: Degenerate (모든 샘플을 Tumor로 예측, Recall=100%, Precision=class prior)

# 7. Detailed Analysis

## 7.1 거리 함수별 성능 비교 (Standard MSE)

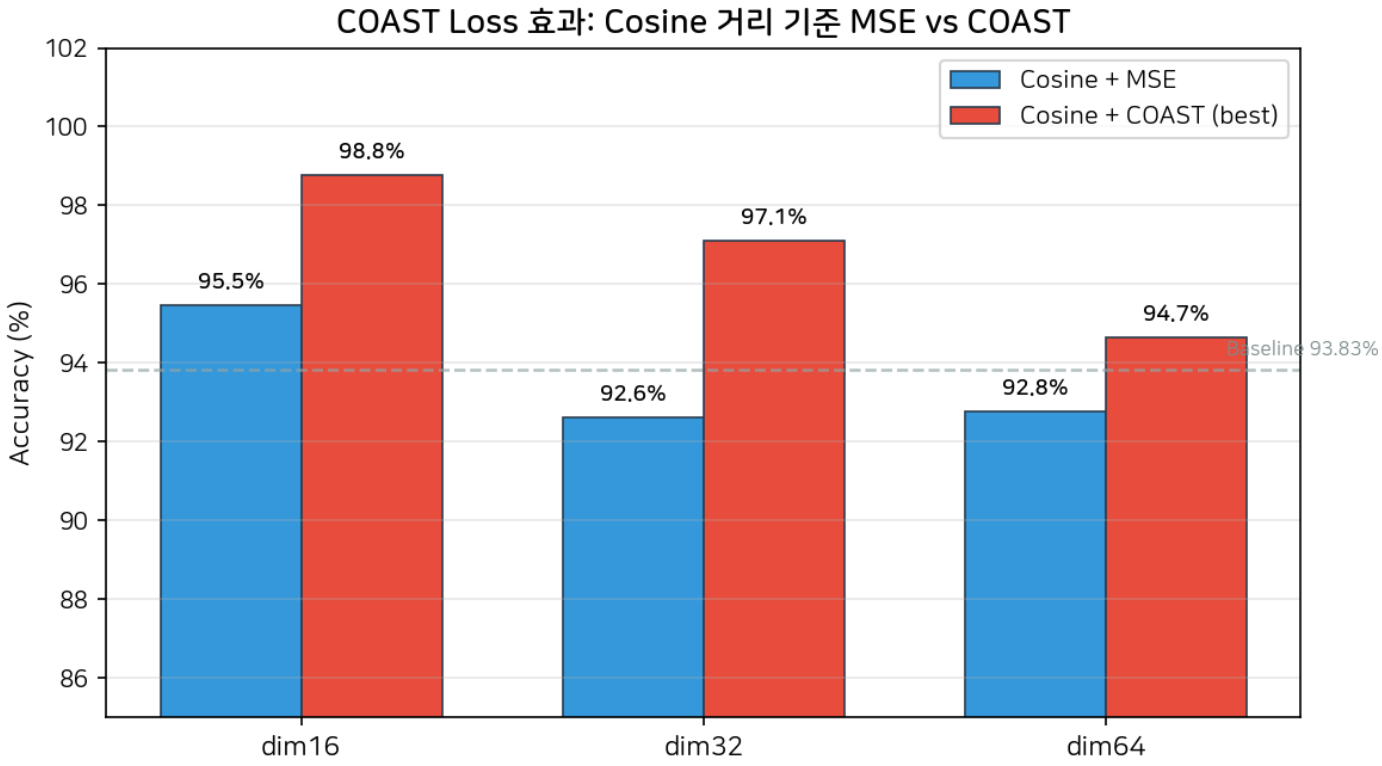



## 7.2 Standard MSE 거리 × 차원 히트맵




90.5% = 퇴화 분류기 (모든 샘플을 Tumor로 예측)

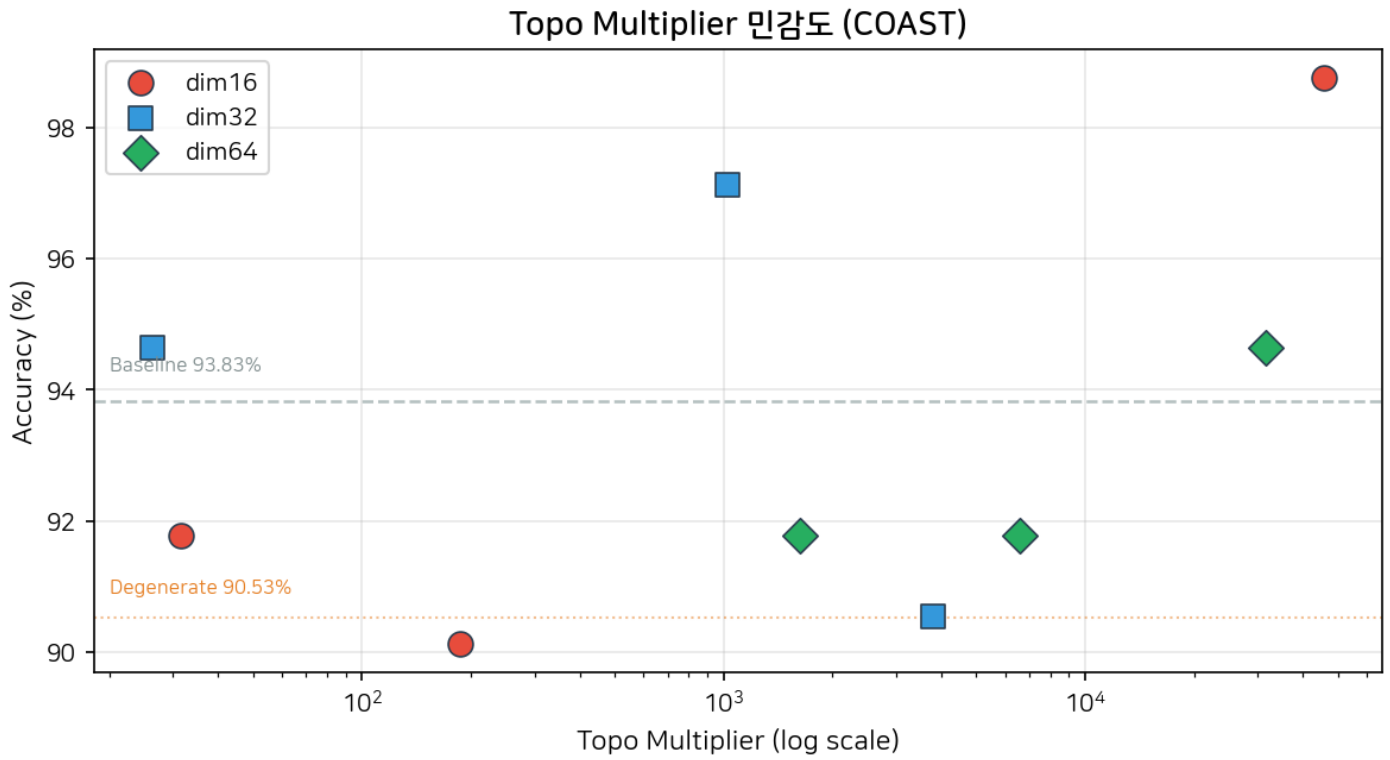
## 7.3 COAST Loss 효과 (Cosine 거리)



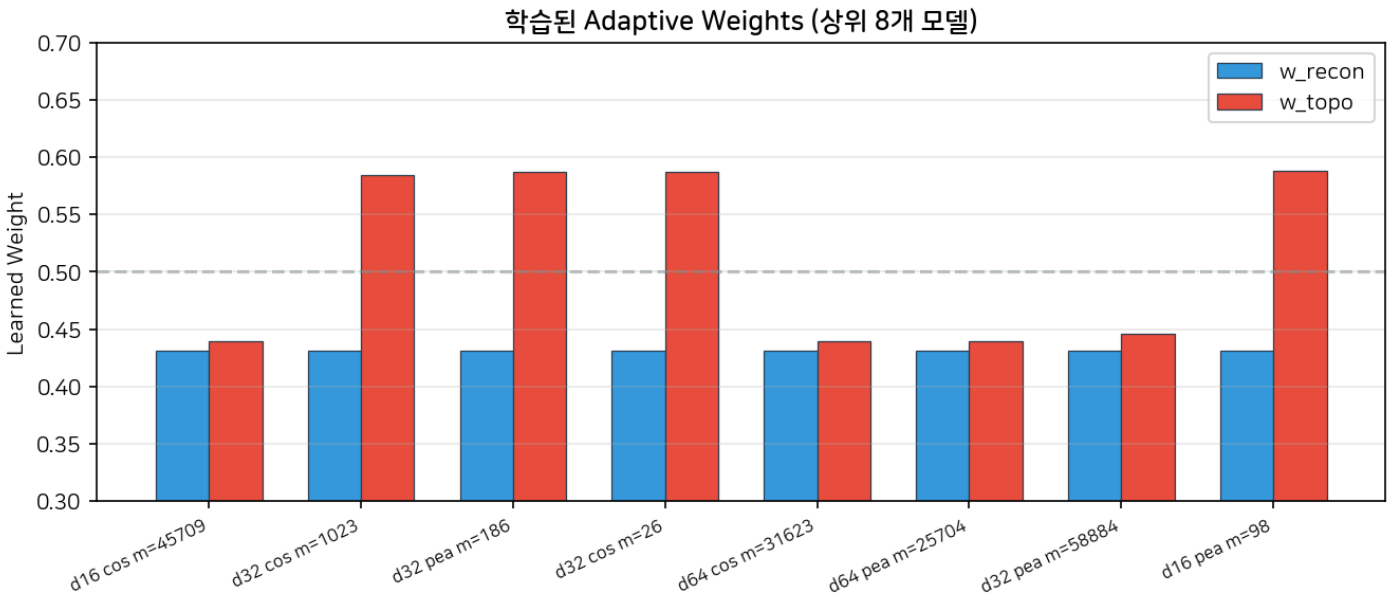

| Dim | Cosine + MSE | COAST (best) | 향상 |
|---|---|---|---|
| 16 | 95.47% | **98.77%** (m=45,709) | +3.30%p |
| 32 | 90.53% * | **97.12%** (m=1,023) | +6.59%p |
| 64 | 90.53% * | **94.65%** (m=31,623) | +4.12%p |

* MSE에서 퇴화했던 dim32, dim64가 COAST로 복구됨

## 7.4 Topo Multiplier 민감도

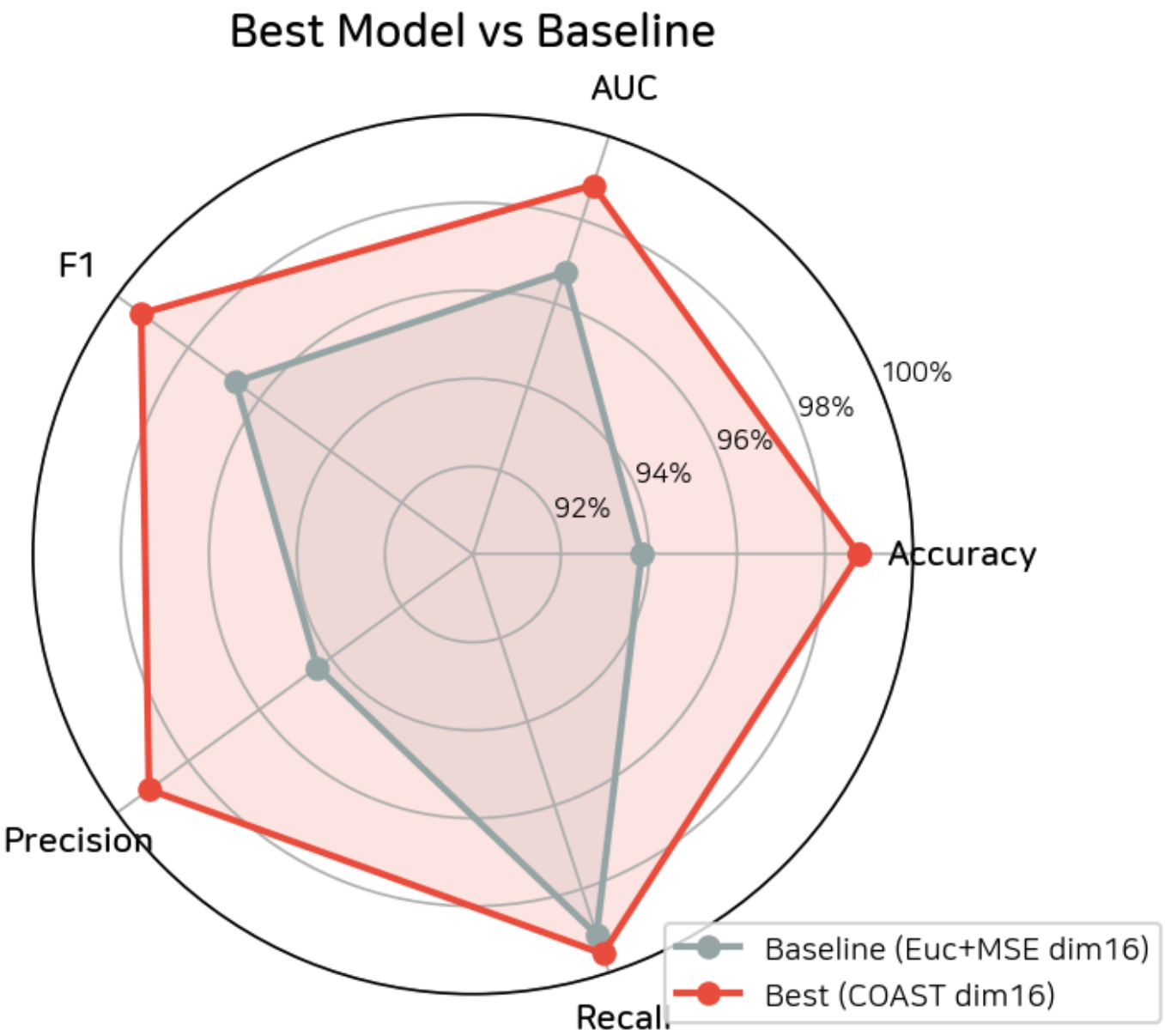



## 7.5 학습된 Adaptive Weights




w_recon ≈ 0.43으로 수렴. 높은 multiplier → w_topo ≈ 0.44 (균형), 낮은 multiplier → w_topo ≈ 0.59 (보상 시도)

## 8. Best Model vs Baseline




| Metric | Baseline (Euc+MSE dim16) | Best (COAST dim16) | Delta |
|---|---|---|---|
| Accuracy | 93.83% | **98.77%** | +4.94%p |
| AUC | 0.968 | **0.988** | +0.020 |
| F1 | 0.967 | **0.993** | +0.026 |
| Precision | 0.944 | **0.991** | +0.047 |
| Recall | 0.991 | **0.996** | +0.005 |
| 압축비 | 20,862 → 16 (1,304 : 1) | | 동일 |

## 9. Key Findings

| # | 발견 | 근거 |
|---|---|---|
| 1 | **Cosine > Euclidean > Pearson** | Std MSE dim16: Cos 95.47% > Euc 93.83% > Prs 90.53% |
| 2 | **COAST가 MSE보다 우수** | 모든 차원에서 COAST가 MSE 대비 +3~6%p 향상 |
| 3 | **dim16이 최적 차원** | 98.77% (dim16) > 97.12% (dim32) > 94.65% (dim64) |
| 4 | **Topo Multiplier가 핵심** | Sweet spot ~1,000–50,000 (Cosine 기준) |
| 5 | **Adaptive Weighting 효과적** | 수렴 시 w_recon ≈ w_topo (자동 균형) |
| 6 | **COAST가 퇴화 분류기 복구** | MSE에서 90.53%로 퇴화 → COAST 97.12% |
| 7 | **선형 분류기만으로 98.77%** | 잠재 공간이 선형 분리 가능 = TAE가 핵심 구조 학습 |